

HUNTSVILLE EXECUTIVE, AL, USA

WMO: 720307

Lat: 34.861N

Lon: 86.557W

Elev: 230

StdP: 98.59

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.5	-15.9	1.0	-1.9	-12.6	1.3	-0.8	10.2	9.4	8.9	5.6	2.3	0	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	35.0	23.3	33.0	22.8	32.4	22.5	25.3	31.0	24.8	30.4	24.1	29.6	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.2	27.6	23.0	18.2	26.7	22.6	17.8	26.4	78.7	31.5	76.2	30.3	73.8	29.8	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.4	-10.8	36.4	2.8	2.0	-12.8	37.8	-14.5	39.0	-16.0	40.1	-18.1	41.5		
(5)			-11.7	25.9	2.6	1.2	-13.6	26.8	-15.1	27.5	-16.5	28.1	-18.4	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	4.8	7.0	10.9	15.8	20.2	24.8	26.0	25.4	22.4	16.1	9.9	6.7	(6)
(7)		DBStd	8.72	5.98	5.70	5.27	4.51	3.92	2.24	2.16	2.61	3.29	4.66	4.75	5.30	(7)
(8)		HDD10.0	559	182	117	54	8	1	0	0	0	0	6	59	132	(8)
(9)		HDD18.3	1826	419	320	235	100	28	0	0	1	7	99	256	360	(9)
(10)		CDD10.0	2705	21	31	82	181	318	444	495	479	373	196	55	31	(10)
(11)		CDD18.3	932	0	2	4	24	87	194	237	221	129	31	3	1	(11)
(12)		CDH23.3	8747	0	6	36	236	827	1870	2241	2049	1162	307	13	1	(12)
(13)		CDH26.7	3572	0	2	2	37	259	814	1003	929	448	76	1	0	(13)
(14)	Wind	WSAvg	2.2	2.8	3.0	3.0	2.7	2.0	1.7	1.5	1.3	1.4	1.9	2.3	2.6	(14)
(15)	Precipitation	PrecAvg	1413	148	136	134	123	107	99	133	92	94	77	106	165	(15)
(16)		PrecMax	1787	287	268	242	251	258	193	272	202	197	183	188	453	(16)
(17)		PrecMin	803	71	68	35	51	16	25	55	25	10	0	43	62	(17)
(18)		PrecStd	222	51	58	52	52	63	54	46	36	55	52	40	86	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	23.7	26.4	29.8	32.7	35.3	37.1	37.3	34.3	31.3	25.3	22.1	(19)
(20)			MCWB	15.0	17.9	17.4	18.4	22.6	21.9	23.5	22.8	22.4	21.5	17.1	16.6	(20)
(21)		2%	DB	17.9	21.0	24.0	27.5	31.1	33.6	34.8	35.1	32.6	28.5	22.4	19.9	(21)
(22)			MCWB	13.9	16.8	16.3	17.7	21.4	22.3	23.7	23.9	21.5	19.8	15.8	15.8	(22)
(23)		5%	DB	17.0	18.5	22.2	26.2	29.8	32.5	32.7	32.8	31.2	27.0	20.4	17.6	(23)
(24)			MCWB	13.4	14.0	14.8	16.8	20.2	22.0	23.3	23.0	21.2	18.4	15.5	14.3	(24)
(25)		10%	DB	14.9	16.2	20.1	24.3	27.7	31.3	32.0	32.0	29.8	24.9	18.1	15.9	(25)
(26)			MCWB	11.9	11.9	14.1	16.0	19.0	21.8	23.1	22.7	21.2	17.8	13.7	12.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	18.9	19.0	20.8	23.9	25.5	26.2	26.1	24.6	23.0	19.4	17.9	(27)
(28)			MCD B	18.5	22.1	24.2	26.7	30.4	31.0	31.4	32.0	29.7	27.9	22.2	20.5	(28)
(29)		2%	WB	15.1	17.5	17.7	19.7	22.7	24.6	25.5	25.2	23.7	22.1	18.3	16.6	(29)
(30)			MCD B	17.3	20.1	22.2	25.3	29.0	30.2	31.0	31.1	29.1	26.5	20.9	18.6	(30)
(31)		5%	WB	14.0	15.4	16.5	18.7	21.8	23.9	24.9	24.6	23.0	20.7	16.7	15.1	(31)
(32)			MCD B	16.6	17.7	20.3	23.7	27.3	29.3	30.4	30.3	28.1	24.1	19.2	17.3	(32)
(33)		10%	WB	12.3	13.1	15.2	17.7	20.7	23.1	24.3	23.9	22.2	19.1	14.9	12.9	(33)
(34)			MCD B	14.2	15.6	18.9	22.0	25.1	28.4	29.6	29.3	26.9	22.8	17.5	15.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.6	11.8	12.8	12.2	11.8	11.0	11.5	12.5	13.1	12.3	9.8	(35)
(36)			MCDBR	11.3	13.1	14.4	14.4	14.0	13.7	12.5	13.3	14.3	15.0	14.1	11.0	(36)
(37)			MCWBR	8.3	8.6	7.7	6.1	5.2	4.3	4.1	4.1	4.9	6.7	8.2	8.1	(37)
(38)		5% WB	MCD BR	9.8	10.9	11.5	11.5	12.0	11.4	10.8	11.4	12.1	11.4	10.0	9.8	(38)
(39)			MCWBR	8.2	8.5	7.0	5.9	5.1	4.5	4.1	4.0	4.9	6.5	7.2	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.325	0.367	0.398	0.449	0.483	0.521	0.531	0.430	0.379	0.339	0.316	(40)	
(41)		taud	2.482	2.466	2.370	2.275	2.179	2.115	2.029	1.990	2.261	2.388	2.446	2.512	(41)	
(42)		Ebn at Noon	886	922	903	887	843	810	776	760	828	847	853	869	(42)	
(43)		Edh at Noon	85	96	115	133	148	158	171	175	127	103	87	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.40	2.93	4.09	5.33	5.85	6.24	5.94	5.50	4.76	3.82	2.78	2.06	(44)	
(45)		RadStd	0.19	0.36	0.39	0.37	0.45	0.48	0.32	0.29	0.42	0.50	0.30	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+98

Nomenclature: See separate page